



Fig. 7: Suggested enclosure



Fig. 8: Another application of the circuit

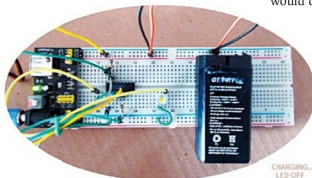


Fig. 9: Author's prototype on a breadboard

LED used in the prototype.

Construction and testing

An actual-size, single-side PCB pattern for the USB LED night light is shown in Fig. 5 and its component layout in Fig. 6. After assembling the circuit on a PCB, enclose it in a suitable box. Connect the circuit to a 5V DC power supply through a USB port.

Alternatively, you can assemble the circuit on a perfboard. If you are designing your own PCB, maintain compactness and adhere to the advice given in the piranha LED datasheet.

lot of vertical space.

Piranha LEDs, also known as superflux LEDs, are widely used for back lighting and illumination. The low-profile package can be easily coupled with reflectors or lenses to efficiently distribute light and provide the desired lit appearance. Fig. 4 depicts the outline drawing and basic specifications of the generic 5mm piranha white

PARTS LIST

Semiconductors:	
IC1	- 74HC4066N quad bilateral switch
D1	- 1N4007 rectifier diode
LED1	- 5mm (4-pin) piranha white LED
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1, R3	- 100-kilo-ohm
R2	- 56-ohm
Capacitors:	
C1	- 0.1 μ F ceramic disk
Miscellaneous:	
CON1	- USB A-type
S1	- SPDT switch
BATT.1	- 4V, 500mAh SMF lead-acid battery
	- 5V, 500mA DC power supply/solar panel

This circuit can be fitted into a bulb-shaped enclosure as shown in Fig. 7. Although a 4V, 500mAh SMF battery is used in the prototype, you can try any other low-profile 4V lead-acid battery, even a type with lesser current rating (< 500mAh) would do.

This night-lamp (Fig. 8) can be used as a portable lighting equipment (torch, headlamp, lantern or tent-light) to make your camping, fishing and other outdoor adventures simpler and more enjoyable. Author's prototype wired on

a breadboard is shown in Fig. 9.

Notes.

1. Remaining switches of IC1 (IC1C and IC1D) can be used to drive more piranha LEDs by making similar connections.

2. Maximum voltage available from the fully-charged battery is about 4.2V, and V_f of the piranha LED used is 3.2V. Based on these observations, value of the series resistor (R2) is set to 56-ohm. **BEY**



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Control with Precision



RE57 Temperature Control

- Universal Measuring Input
- Digital Input Filter - Timer settable from 0 ... 999.9 seconds
- Control Type: PID and ON / OFF
- Onsite Mode Selection: Auto / Manual
- Timer Function: with 5 functional selection
- Onsite programmability - Heating / cooling
- Output on Failure function



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